Operator Action SOP

Standard Operating Procedure for AI-assisted quality inspection using selected MVTec AD categories

Field	Details
Document ID	OAS-MVTEC-AD-001
Version	1.0
Prepared for	RAG knowledge base, AI agent routing, inspection UI, and report generation
Covered categories	bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut
Related documents	QA Inspection Manual, Defect Severity Guide, Root-Cause Troubleshooting Guide, Inspection Report Template
Primary users	Quality inspector, line operator, supervisor, AI application developer, student project evaluator
Document type	Synthetic standard operating procedure for prototype/student use

Important note

This SOP is a simulated but realistic operator action guide created for a prototype RAG system. It is not an official MVTec document, not a certified manufacturing standard, and not a replacement for a real factory's approved quality-control procedures. The AI assistant should use this SOP as supporting evidence and must route uncertain, critical, or safety-related cases to a human inspector.

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1. Purpose and Scope

The purpose of this Operator Action SOP is to define what a quality-control operator should do after the AI Defect Detection and Root-Cause Assistant returns a result. The document converts technical outputs such as anomaly score, confidence, heatmap, defect category, severity level, and retrieved evidence into simple operational actions such as accept, rework, reject, quarantine, retake image, or escalate to supervisor.

This SOP supports the RAG module by providing clear action rules. When the user asks what should be done next, the retriever should find the relevant action passage from this document and pass it to the LLM. The LLM must then explain the action in simple language and avoid unsupported claims.

The SOP applies to the selected MVTec AD prototype categories: bottle, cable, screw, transistor, tile, toothbrush, zipper, and metal_nut. It is designed for image-based inspection using uploaded product images or video frames. It may also be used with OCR output when the product image contains labels, serial numbers, batch IDs, or printed codes.

1.1 What this SOP covers

- What the operator should do when the product is predicted as normal.
- What the operator should do when the defect is minor, major, or critical.
- When the operator should request another image instead of trusting the prediction.
- When the product should be reworked, rejected, quarantined, or escalated.
- How to handle confidence scores, OCR uncertainty, missing RAG evidence, and unclear heatmaps.
- How to document the result in the final inspection report.

1.2 What this SOP does not cover

- It does not provide exact factory-specific tolerances or legal compliance rules.
- It does not prove the true root cause of a defect from an image alone.
- It does not replace final approval by a certified quality engineer or supervisor.
- It does not define machine repair instructions beyond first-level inspection and escalation actions.

2. Operator Roles and Responsibility Levels

The AI assistant should support different users in the inspection process. Each user has a different responsibility level. The system should not allow the LLM to appear as the final authority when the decision requires human judgement.

Role	Responsibility
Line operator	Uploads image, checks prediction, follows simple action instructions, separates suspected defective items, and records visible issues.
Quality inspector	Reviews AI output, verifies severity, confirms accept/rework/reject decisions, and checks retrieved SOP evidence.
Supervisor	Approves critical rejects, quarantine decisions, repeated defect trends, and process stoppage decisions.
Process engineer	Investigates repeated root-cause patterns, machine setting issues, tooling wear, alignment problems, contamination sources, or material issues.
AI system	Provides prediction, confidence, heatmap or mask, retrieved evidence, recommendation, and draft report. It does not replace human accountability.

2.1 Responsibility rule

The AI assistant may recommend an action, but the human operator or quality inspector is responsible for final release, rejection, or escalation. If the model confidence is low, evidence is missing, or the defect is critical, the AI must route the case to human review.

3. Standard AI Inspection Workflow

The operator should follow the same workflow for every uploaded image. A consistent workflow improves traceability, makes reports easier to audit, and reduces the chance that a serious defect is missed.

Step	Action	Operator instruction
1	Capture or upload image	Take a clear image showing the full product. Avoid motion blur, strong glare, heavy shadow, and partial crop.
2	Run image quality check	If the image is blurry, dark, too bright, too small, or incomplete, retake image before prediction.
3	Run vision inference	Model predicts normal/defective and may return category, anomaly score, heatmap, mask, and confidence.
4	Review visual evidence	Operator checks whether the highlighted region matches a real visible defect.
5	Check severity	Use Defect Severity Guide to map model output and visual evidence to normal, minor, major, critical, or human review.
6	Retrieve SOP evidence	RAG retrieves action rule from this SOP and related manuals before the LLM writes a recommendation.
7	Take action	Accept, rework, reject, quarantine, retake image, or escalate based on confidence and severity.
8	Complete report	Save inspection ID, image name, category, model output, severity, evidence, action, reviewer, and timestamp.

3.1 Required system behavior

- The UI should show the uploaded image and the model output in one result screen.
- The UI should show a clear action label such as ACCEPT, REWORK, REJECT, QUARANTINE, RETAKE IMAGE, or HUMAN REVIEW.
- The recommendation must include the evidence source retrieved by RAG whenever the user asks for an explanation or next action.
- The report generator must store the final action and whether a human confirmed the decision.

4. Action Codes Used by the System

The AI assistant should use consistent action codes. These action codes can be displayed in the frontend, stored in the database, and written into inspection reports.

Action code	Meaning	Operator action	Typical trigger
ACCEPT	Product appears normal and confidence is sufficient	Allow product to continue	Normal prediction; no visible anomaly; image quality acceptable
ACCEPT_WITH_NOTE	Very minor cosmetic issue only	Continue product but record note	Small mark or harmless variation; no function or safety impact
REWORK	Defect may be corrected	Move item to rework area	Minor to moderate defect that can be cleaned, adjusted, trimmed, polished, reassembled, or reinspected
REJECT	Product should not be used or shipped	Remove item from accepted output	Major defect affecting function, fit, safety, sealing, electrical behavior, hygiene, or customer acceptance
QUARANTINE	Batch or group requires hold	Separate product/batch until supervisor review	Repeated defects, possible contamination, uncertain batch label, critical defect pattern, or process concern
RETAKE_IMAGE	Input image is not reliable	Capture clearer image	Blur, poor lighting, partial product, low resolution, glare, shadow, wrong view
HUMAN_REVIEW	AI cannot decide confidently	Route to inspector/supervisor	Low confidence, missing evidence, conflicting outputs, critical severity, OCR mismatch, or ambiguous heatmap
ESCALATE_PROCESS	Possible process-level issue	Notify supervisor/engineer	Repeated same defect, critical defect, high reject rate, or likely machine/tooling issue

5. Confidence, Severity, and Human Review Rules

The operator action must not depend only on the predicted label. It should consider confidence score, image quality, severity level, OCR reliability, and whether RAG found a relevant action rule.

Condition	Interpretation	Required action
>= 90%	High confidence	Follow severity-based action, but critical cases still require supervisor review.
70% to 89%	Medium confidence	Allow recommendation for minor/major cases, but show caution and allow human confirmation.
50% to 69%	Low confidence	Do not make final decision automatically. Route to HUMAN_REVIEW unless defect is clearly visible.
< 50%	Very low confidence	Treat as uncertain. Request RETAKE_IMAGE or HUMAN_REVIEW.
Image quality failed	Unreliable input	Use RETAKE_IMAGE before any accept/reject decision.
No RAG evidence	Unsupported recommendation	Answer with caution and route to HUMAN_REVIEW for action decision.

5.1 Severity-based action mapping

Severity	Default action	Notes
Normal	ACCEPT	Only if image quality is acceptable and no visible anomaly is highlighted.
Minor	ACCEPT_WITH_NOTE or REWORK	Use rework if cosmetic defect can be corrected or customer appearance standard is strict.
Major	REWORK or REJECT	Reject when defect affects function, fit, sealing, electrical behavior, assembly, structural integrity, or hygiene.
Critical	REJECT and ESCALATE_PROCESS	Supervisor review required. Quarantine may be required if batch-level risk exists.
Uncertain	RETAKE_IMAGE or HUMAN_REVIEW	Do not claim final action if model evidence or document evidence is weak.

5.2 Mandatory human review triggers

- Model confidence is below 70% for a defect-related decision.
- Severity is critical or safety-related.
- The image quality check fails or the product is not fully visible.
- The heatmap or mask highlights background instead of the product.
- OCR reads a batch number, serial number, or label that conflicts with operator input.
- RAG cannot retrieve a relevant SOP passage for the recommended action.
- The same defect repeats across multiple images, suggesting a process-level issue.
- The operator disagrees with the AI output after visual inspection.

6. General Operator Actions for All Categories

These general actions apply to all selected product categories. Category-specific rules are provided in Section 7.

Situation	Operator check	Action	Documentation note
Normal prediction	Check that image quality passed and no visible defect is present	ACCEPT	Store result for traceability
Minor cosmetic anomaly	Compare with severity guide and check whether function is affected	ACCEPT_WITH_NOTE or REWORK	Use rework if defect is visible to customer
Major visible defect	Separate product and check category-specific SOP	REWORK or REJECT	Do not release until human confirms
Critical defect	Remove product from flow immediately	REJECT, QUARANTINE, ESCALATE_PROCESS	Supervisor approval required
Unclear image	Retake image with correct lighting and full product view	RETAKE_IMAGE	Do not run final action on poor image

Situation	Operator check	Action	Documentation note
Repeated defect trend	Hold related batch or sample more items	QUARANTINE and ESCALATE_PROCESS	Possible machine or process issue
Unsupported AI answer	Check whether RAG evidence was retrieved	HUMAN_REVIEW	LLM must not invent action

6.1 Image retake instructions

- Place product on a plain background where the object is clearly separated from surroundings.
- Use stable lighting and avoid extreme glare or deep shadow.
- Capture the full product and the suspected defect area in focus.
- For reflective objects, adjust angle to reduce reflection while keeping the defect visible.
- For small items such as screw and metal nut, use closer view without cropping edges.
- For cable, zipper, and toothbrush, show the full length when possible and add close-up view for the defect area.

7. Category-Specific Operator SOPs

The following SOP tables define the recommended operator action for each selected MVTec AD category. These rules are written so the RAG retriever can find direct evidence for category-specific action questions.

7.1 Bottle Operator Action SOP

Category keyword for RAG: bottle. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Visible crack, broken rim, broken neck, or chipped sealing surface	Critical	REJECT and ESCALATE_PROCESS	Do not release. Separate item. Check handling, molding, impact, and transport conditions.
Contamination inside bottle or foreign material on inner wall	Critical	REJECT and QUARANTINE	Hold related batch if repeated. Inspect cleaning, storage, and environmental controls.
Surface stain or dirt on outside only	Minor to Major	REWORK or REJECT	Clean and reinspect if external only. Reject if contamination cannot be removed or affects customer acceptance.
Deformation, dent, uneven shape, or unstable standing surface	Major	REJECT or HUMAN_REVIEW	Check forming process and handling. Do not accept if sealing or stability may be affected.
Low confidence anomaly on background or reflection	Uncertain	RETAKE_IMAGE	Use better lighting and angle. Do not reject only from glare.

Mandatory review rule for bottle: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

7.2 Cable Operator Action SOP

Category keyword for RAG: cable. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Cut, exposed conductor, insulation rupture, or deep tear	Critical	REJECT and ESCALATE_PROCESS	Potential electrical/safety risk. Remove item and notify quality supervisor.
Bent, twisted, kinked, or crushed cable section	Major	REJECT or HUMAN_REVIEW	Check whether shape affects connection, strength, or installation. Escalate if repeated.
Connector damage, missing connector part, or deformed end	Critical	REJECT	Do not rework unless approved by engineering. Verify connector assembly process.
Small dirt mark or removable surface contamination	Minor	REWORK	Clean item and reinspect. Reject if contamination remains or covers critical area.
Unclear anomaly due to cable crossing or background	Uncertain	RETAKE_IMAGE	Capture cable on plain background with full length visible.

Mandatory review rule for cable: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

7.3 Screw Operator Action SOP

Category keyword for RAG: screw. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Broken head, missing head section, or severe deformation	Critical	REJECT	Product cannot perform fastening function. Check feed, handling, and tooling.
Damaged thread, stripped thread, bent shaft, or major surface burr	Major to Critical	REJECT and HUMAN_REVIEW	Reject if assembly fit is affected. Escalate if many screws show same thread defect.
Minor discoloration or small cosmetic surface mark	Minor	ACCEPT_WITH_NOTE or REWORK	Accept only if no corrosion, burr, or thread damage is present.
Rust, corrosion-like stain, or embedded contamination	Major	REJECT or QUARANTINE	Hold batch if corrosion/contamination appears repeated.
Model heatmap uncertain due to small size or blur	Uncertain	RETAKE_IMAGE	Use close-up image with screw fully in focus.

Mandatory review rule for screw: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

7.4 Transistor Operator Action SOP

Category keyword for RAG: transistor. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Bent, broken, missing, or misaligned pin/lead	Critical	REJECT and ESCALATE_PROCESS	Electrical assembly may fail. Check placement, transport, and insertion process.
Cracked package body or broken casing	Critical	REJECT	Do not rework. Possible reliability and safety risk.
Contamination, solder residue, or foreign material near leads	Major	REWORK or REJECT	Clean if allowed and reinspect. Reject if residue cannot be removed or affects connection.
Minor printing/marking issue without structural damage	Minor	HUMAN_REVIEW or ACCEPT_WITH_NOTE	Verify product identification if OCR/label text matters.
Low confidence anomaly on shadow or reflection	Uncertain	RETAKE_IMAGE	Capture with uniform lighting and clear lead visibility.

Mandatory review rule for transistor: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

7.5 Tile Operator Action SOP

Category keyword for RAG: tile. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Crack, broken edge, missing corner, or deep fracture	Critical	REJECT	Structural and customer acceptance issue. Remove item from accepted output.
Surface chip, deep scratch, or visible pit	Major	REJECT or REWORK	Rework only if polishing/repair is approved and defect is not structural.
Color variation, stain, or surface mark	Minor to Major	REWORK or HUMAN_REVIEW	Clean/rework if removable. Reject if permanent or outside visual standard.
Pattern variation within acceptable design range	Normal or Minor	ACCEPT_WITH_NOTE	Accept only if defect guide confirms it is acceptable natural variation.
Image affected by lighting gradient	Uncertain	RETAKE_IMAGE	Use flat lighting and full tile view.

Mandatory review rule for tile: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

7.6 Toothbrush Operator Action SOP

Category keyword for RAG: toothbrush. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Broken handle, cracked neck, sharp edge, or detached part	Critical	REJECT	Possible user injury or product failure. Escalate if repeated.
Missing, damaged, contaminated, or deformed bristles	Major to Critical	REJECT or QUARANTINE	Hygiene and usability issue. Quarantine if contamination may affect batch.
Minor cosmetic discoloration on handle	Minor	REWORK or ACCEPT_WITH_NOTE	Clean/rework if allowed. Reject if stain remains or suggests contamination.
Packaging/label text issue if visible	Major	HUMAN_REVIEW	Use OCR verification when label or batch code matters.
Blur on bristle region	Uncertain	RETAKE_IMAGE	Capture close-up with bristles clearly visible.

Mandatory review rule for toothbrush: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

7.7 Zipper Operator Action SOP

Category keyword for RAG: zipper. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Missing, broken, or severely misaligned teeth	Critical	REJECT	Zipper may not close correctly. Remove product from accepted output.
Damaged slider, stuck slider, or missing puller	Critical	REJECT and ESCALATE_PROCESS	Functional failure. Check assembly process if repeated.
Fabric tape tear, stitching defect, or frayed edge	Major	REWORK or REJECT	Rework if repair is allowed and strength is not affected. Otherwise reject.
Small cosmetic mark on tape	Minor	ACCEPT_WITH_NOTE or REWORK	Clean/rework and reinspect if visible to customer.
Heatmap unclear due to folded zipper	Uncertain	RETAKE_IMAGE	Lay zipper flat and capture full length plus close-up.

Mandatory review rule for zipper: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

7.8 Metal Nut Operator Action SOP

Category keyword for RAG: metal_nut. The operator should use the following actions after the model predicts an anomaly in this category.

Detected condition	Severity	Required action	Operator note
Cracked body, broken edge, or severe deformation	Critical	REJECT and ESCALATE_PROCESS	Cannot guarantee strength or fit. Check machining, forming, or handling process.
Damaged internal thread, missing thread area, or heavy burr	Critical	REJECT	Functional failure risk. Do not release.
Surface scratch, small dent, or shallow mark	Minor to Major	ACCEPT_WITH_NOTE, REWORK, or REJECT	Decision depends on depth, location, and fit impact. Rework if polish/deburr is allowed.
Rust, corrosion, oil residue, or embedded contamination	Major	REWORK, REJECT, or QUARANTINE	Clean if removable. Quarantine if repeated or corrosion-like.
Low confidence due to reflection on metal surface	Uncertain	RETAKE_IMAGE	Change angle/lighting and capture thread region clearly.

Mandatory review rule for metal_nut: if the model confidence is below 70%, if the highlighted area does not match a visible defect, or if the image does not show the full product, route the inspection to HUMAN_REVIEW or RETAKE_IMAGE instead of making a final accept/reject decision.

8. Rework, Reject, Quarantine, and Escalation Rules

This section defines how the operator should handle the physical product after the recommended action is selected. The goal is to prevent defective or uncertain items from being mixed with accepted items.

Action	Physical handling	Examples	Control rule
Rework	Move item to controlled rework area; record defect and rework type; reinspect after correction	Clean, polish, deburr, adjust, trim, straighten, relabel, or reassemble only if allowed	Reworked product must pass a second inspection
Reject	Remove from accepted flow; mark as rejected; store in reject container	Cracks, broken structures, missing functional parts, unsafe defects, failed assembly features	Reject reason must be recorded
Quarantine	Hold item or batch separately until supervisor decision	Repeated defects, possible contamination, label mismatch, batch-level risk, uncertain critical issue	Do not release quarantined item without approval
Escalate	Notify supervisor/process engineer; attach image and report	Critical defect, repeated pattern, sudden defect increase, possible machine/process issue	Escalation ID should be linked to inspection report

8.1 Rework acceptance rule

A reworked item must not be automatically accepted. It must be reinspected after correction. The report should contain both the original defect result and the reinspection result. If the same defect remains after rework, the item should be rejected or routed to human review.

8.2 Batch quarantine rule

If the same major or critical defect appears repeatedly in the same category, shift, batch, or machine line, the operator should quarantine related items and escalate to the supervisor. The AI system should suggest a process-level review but should not claim the final root cause without additional data.

9. OCR and Label Verification Actions

OCR is optional in the first version of the project, but it becomes useful when the product image contains serial numbers, batch codes, expiry dates, printed markings, or product labels. OCR output should be treated as supporting information, not as perfect truth.

OCR situation	Operator check	Action
OCR reads label clearly	Compare OCR text with expected product/category/batch data	Record OCR output in report
OCR confidence low	Do not use OCR result for final decision	HUMAN_REVIEW or manual entry
Batch code unreadable	Request close-up image of label or manual verification	RETAKE_IMAGE
OCR text conflicts with operator input	Hold item until verified	HUMAN_REVIEW
Wrong or missing label on product/package	Treat as major documentation/traceability issue	REJECT or QUARANTINE depending on policy

OCR evidence should be written in the report as 'OCR-read text' rather than 'confirmed text' unless a human verified it.

10. RAG Evidence Requirements Before Answering

When the user asks for the next action, the AI assistant should retrieve evidence from SOP or QA documents before producing the final answer. The LLM should not invent actions from general knowledge when a document-based decision is expected.

User request	Required retrieval	LLM output rule
Question asks what to do next	Retrieve Operator Action SOP and Defect Severity Guide	Final answer must include action and reason
Question asks why defect happened	Retrieve Root-Cause Troubleshooting Guide	Use possible cause language only

User request	Required retrieval	LLM output rule
Question asks if product can pass	Retrieve QA Inspection Manual and Severity Guide	Answer with accept/rework/reject/human review
Question asks for report	Retrieve report template and action SOP	Generate structured report
No relevant evidence found	Do not pretend evidence exists	Say evidence is insufficient and route to human review

10.1 Evidence statement format

The final answer should include a short evidence statement such as: 'Based on the Operator Action SOP, a cable with exposed conductor should be rejected and escalated because it is a critical safety-related defect.' If the evidence is weak or missing, the final answer should say: 'The system did not retrieve enough SOP evidence to make a final action decision. Human review is required.'

11. Report Completion Rules

Every inspection should produce a structured record. This is important for traceability, debugging, evaluation, and future improvement of the AI system.

Report field	Description	Status
Inspection ID	Unique ID generated by system	Required
Date and time	Timestamp of inspection	Required
Product category	bottle, cable, screw, transistor, tile, toothbrush, zipper, or metal_nut	Required
Image filename	Original uploaded file name or stored path	Required
Model output	Normal/defective, confidence, anomaly score, heatmap/mask if available	Required
Severity	Normal, minor, major, critical, or human review required	Required
Recommended action	ACCEPT, REWORK, REJECT, QUARANTINE, RETAKE_IMAGE, HUMAN_REVIEW, ESCALATE_PROCESS	Required
Retrieved evidence	Relevant section from SOP/manual	Required for explained answer
Human confirmation	Inspector name or confirmation status	Required for final approval
Notes	Operator comments, rework details, escalation notes	Optional but recommended

11.1 Report wording rules

- Use 'possible root cause' instead of 'definite root cause' unless supported by direct process data.
- Use 'recommended action' instead of 'final decision' until human confirmation is recorded.
- Mention low confidence clearly if the model confidence is below the configured threshold.
- Include retrieved evidence section when the report uses RAG output.
- Do not hide uncertainty. If evidence is missing, write that human review is required.

12. Safety, Ethics, and Limitations

This SOP is designed for decision support, not fully automated industrial release. The system should be transparent about confidence, evidence, and uncertainty.

- Do not use the AI system as the only authority for safety-critical product release.
- Do not allow the LLM to make unsupported claims about root cause or product safety.
- Do not auto-accept items when the image quality check fails.
- Do not auto-reject products only because of glare, shadow, background artifact, or low-quality image.
- Maintain traceability by saving images, predictions, evidence snippets, and human confirmation.

- Review model performance regularly using false positive and false negative cases.

13. RAG Retrieval Keywords and Expected Evidence

The following keyword groups help the RAG system retrieve the correct section. Developers can use these terms for testing retrieval quality.

Intent	Retrieval keywords
Accept/pass	normal product, no anomaly, acceptable variation, pass inspection, ACCEPT
Rework	clean, polish, deburr, adjust, repair, reinspect, REWORK
Reject	crack, broken, missing part, functional failure, unsafe, REJECT
Quarantine	batch hold, repeated defect, contamination, traceability issue, QUARANTINE
Escalate	critical defect, repeated pattern, supervisor, process engineer, ESCALATE_PROCESS
Image retake	blurry, dark, cropped, glare, shadow, low resolution, RETAKE_IMAGE
Human review	low confidence, uncertain, missing evidence, conflicting output, HUMAN_REVIEW
Category terms	bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut

14. Example Operator Scenarios

Scenario	Severity	Recommended action	Reason
Bottle crack detected with 94% confidence	Critical	REJECT and ESCALATE_PROCESS	Visible crack may affect sealing/strength. Supervisor review required.
Cable surface dirt with 82% confidence	Minor	REWORK	Clean and reinspect. Reject if contamination remains or enters critical area.
Screw thread defect with 88% confidence	Major/Critical	REJECT or HUMAN_REVIEW	Thread damage may affect fastening function.
Transistor bent lead with 93% confidence	Critical	REJECT	Electrical assembly failure risk.
Tile lighting gradient highlighted as anomaly	Uncertain	RETAKE_IMAGE	Lighting issue should not be treated as true defect.
Toothbrush bristle contamination	Major/Critical	REJECT or QUARANTINE	Hygiene risk; batch review if repeated.
Zipper missing teeth	Critical	REJECT	Functional closure failure.
Metal nut reflective glare with 55% confidence	Uncertain	RETAKE_IMAGE	Need better image before action.

15. RAG Chunking Guidance

For best retrieval, this SOP should be chunked by section and category. Each chunk should preserve the section title, category name, action code, severity, and operator instruction. Avoid splitting a single category table row across different chunks because the model needs condition, severity, action, and operator note together.

Chunk type	Recommended content	Why it matters
Global workflow chunk	Sections 3-6	Helps answer general questions such as 'what should the operator do first?'
Action code chunk	Section 4	Helps map system labels to user-friendly actions.
Category chunk	Each category in Section 7	Helps retrieve precise action for bottle/cable/screw/etc.
Report chunk	Section 11	Helps generate consistent inspection reports.
Safety chunk	Section 12	Helps prevent overconfident or unsafe answers.

15.1 Metadata fields for vector database

- document_name: Operator_Action_SOP
- document_type: SOP

- category: bottle/cable/screw/transistor/tile/toothbrush/zipper/metal_nut/general
- action_code: ACCEPT/REWORK/REJECT/QUARANTINE/RETAKE_IMAGE/HUMAN_REVIEW/ESCALATE_PROCESS
- severity: normal/minor/major/critical/uncertain
- version: 1.0

16. Sample Questions for Testing Retrieval

Use these questions to test whether the RAG retriever can find the correct action evidence from this SOP.

1. What should the operator do if a bottle has a visible crack?
2. When should a cable defect be rejected instead of reworked?
3. What action is required for a screw with damaged thread?
4. How should the system handle a transistor with a bent lead?
5. Should a tile with a lighting artifact be rejected?
6. What should the operator do when toothbrush bristles look contaminated?
7. What is the action for a zipper with missing teeth?
8. How should a metal nut with possible glare and low confidence be handled?
9. When should the AI ask for human review?
10. What fields must be included in the inspection report?

End of document

This Operator Action SOP is prepared as a synthetic RAG knowledge document for the AI Defect Detection and Root-Cause Assistant prototype. It should be used together with the QA Inspection Manual, Defect Severity Guide, Root-Cause Troubleshooting Guide, and Inspection Report Template.